Aswin Kumar

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Portfolio: aswinkumar-4portfolio.vercel.app

PROFESSIONAL SUMMARY

AI Engineer building Generative AI, RAG, AI agent, machine learning and computer vision applications for real business problems. Built a LangGraph RAG agent with tool calling and LangSmith tracing, a YOLOv8 detector served via FastAPI and Docker, and LLM automations (Claude, Playwright, n8n) that updated 700+ live Amazon listings and Shopify tags. Strong in Python, prompt engineering, model evaluation and cost-aware LLM design.

TECHNICAL SKILLS

Programming and Data: Python, SQL, Pandas, NumPy, Matplotlib, Seaborn

Generative AI and LLMs: LLMs, prompt engineering, RAG, embeddings, vector search, semantic search, NLP, chunking, structured outputs, LLM cost optimisation, Claude, Groq, sentence-transformers

Agents and Orchestration: LangChain, LangGraph, LangSmith, AI agents, agentic workflows, tool calling, state management, guardrails

Machine Learning: scikit-learn, logistic regression, decision trees, transfer learning, model evaluation

Deep Learning and Computer Vision: PyTorch, TensorFlow, Keras, CNNs, YOLOv8, image annotation, data augmentation, object detection, Albumentations, Roboflow, OpenCV, mAP, precision and recall

Backend and Deployment: FastAPI, REST APIs, Pydantic, Uvicorn, Docker, Docker Compose, Git, GitHub

Automation: n8n, Playwright, workflow automation, browser automation, Shopify, Amazon Seller Central

PROJECTS

Hazardous Waste Detection System Freelance | 2025–2026
Client: TMT Manufacturing Company *Python, PyTorch, YOLOv8, OpenCV, FastAPI, Docker*

- Built an end-to-end computer vision system that detects gas cylinders and shock absorbers in cluttered scrap-yard images, so hazardous items can be found before they are crushed or shredded.
- Annotated 98 images in Roboflow, balanced classes with label-aware Albumentations augmentation (1,161 objects) and fine-tuned YOLOv8n from COCO weights with early stopping to 0.92 mAP@50–95 on the validation split.
- Served the 6 MB model through a FastAPI REST API (Swagger docs, singleton model loader) with an HTML/JS upload and webcam UI, containerised with Docker Compose; trained and runs on CPU only.

HR Policy Assistant Freelance | 2025–2026
Client: Laxmi Exim *LangGraph, LangChain, Groq, RAG, LangSmith*

- Built a conversational LangGraph AI agent that answers employee questions from a 57-page HR handbook with RAG (recursive chunking, local all-MiniLM-L6-v2 embeddings, top-4 retrieval) on Groq-hosted gpt-oss-120b.
- Designed a router-first graph (6 nodes, 3 routes) with Pydantic structured output: off-topic questions get a fixed refusal with no LLM call, and policy answers must come from retrieved handbook text.
- Implemented 7 deterministic Python tools via tool calling, so leave, payroll and reimbursement calculations come from code with handbook chapter references; boundary-tested every rule.
- Prevented agent loops with a state flag and tool_choice="none", and traced every run in LangSmith.

PROFESSIONAL EXPERIENCE

Operations Analyst 2026 – Present
Lakshmi Group of Companies *Coimbatore, India*

- Built a Claude and Playwright pipeline that reads each Amazon listing, rewrites its title, description and bullet points for SEO, and publishes the update in Seller Central; applied to 700+ listings, none edited by hand.
- Built an n8n workflow that turns an Excel sheet of SKUs into Shopify tag updates: matches each SKU, uses an LLM only where tagging needs judgment, and returns an Updated / Failed / Unmatched report.
- Analyse sales data and product performance across business units and prepare operational reports for management.
- Identify repetitive workflows across business units for automation and process improvement.

EDUCATION

Bachelor of Computer Applications (BCA) 2022 – 2025
RVS College of Arts & Science *Coimbatore, India*

LANGUAGES

Tamil (Native), English